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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END AND BACKLOG SUBJECT EXAMINATIONS - SEPTEMBER / OCTOBER 2023

Program	: **B.E. – Common to all Programs**	**Semester**	: **I / II**
Course Name	: **Introduction to Civil Engineering**	**Max. Marks**	: **100**
Course Code	: **ESC131 / ESC231**	**Duration**	: **3 Hrs**

Instructions to the Candidates:
- Answer one full question from each unit.
- Assume missing data suitably.

UNIT - I

1. a) Discuss the scope of following fields of Civil Engineering. CO1 (08)
   i) Geotechnical Engineering
   ii) Environmental Engineering.
   b) State advantages and disadvantages of stone masonry construction over brick masonry construction. CO1 (08)
   c) List any four qualities of a good bricks. CO1 (04)

2. a) Explain, with a neat sketch the components of dog legged staircase. CO1 (08)
   b) Discuss the advantages and disadvantages of Prestressed concrete. CO1 (08)
   c) Explain the role of civil engineer in Infrastructure development. CO1 (04)

UNIT - II

3. a) Describe on the impact of infrastructural development on the socio economic development of a country. CO2 (08)
   b) Describe on the concept of sustainability and explain the role of civil engineering in achieving the sustainability. CO2 (06)
   c) Explain any two water management systems in detail. CO2 (06)

4. a) Describe the following: CO2 (08)
   (i) Energy Efficient buildings (ii) Water Supply and Sanitary systems.
   b) Describe: (i) Smart Buildings (ii) Temperature control in buildings. CO2 (06)
   c) Elaborate on : (i) Clean city concept and (ii) Identification of land fill sites CO2 (06)

UNIT - III

5. a) Explain the different types of force systems giving an example for each one of them. CO3 (06)
   b) A system of forces acting are shown in Fig. 5(b). Determine the magnitude and direction of the resultant. CO3 (06)

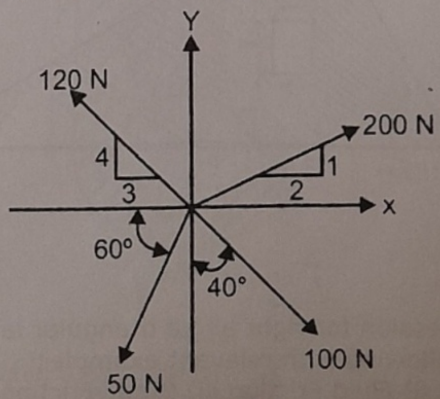

Fig. 5(b)

c) Determine the magnitude and position of the resultant of the forces acting on a body as shown in Fig. 5(c) with respect to point A. CO3 (08)

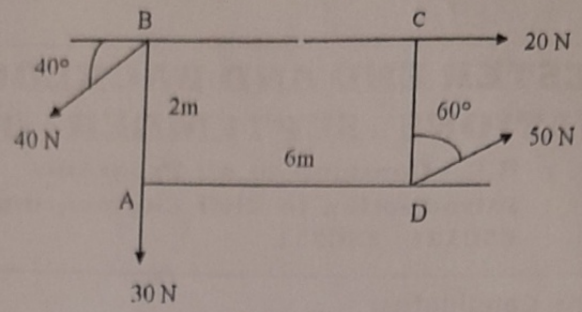

Fig.5(c)

6. a) The sum of two concurrent forces P and Q is 500N and their resultant is 400N. If the resultant is perpendicular to P, determine the value of P and Q. Also determine the angle between P and Q. CO3 (10)

b) A flat plate is subjected to the coplanar system of forces as shown in Fig. 6(b). Each square of the inscribed grid is having a length of 1 m. Determine the magnitude, direction and also the position of the resultant force with respect to point A. CO3 (10)

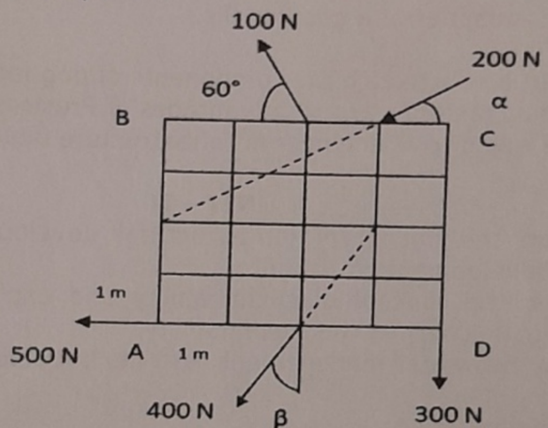

Fig. 6(b)

UNIT- IV

7. a) Determine the centroid for the plane lamina shown in Fig. 7(a). CO4 (10)

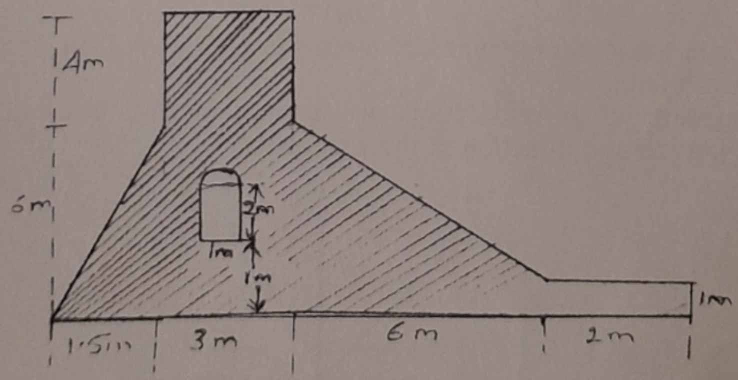

Figure 7a

b) Derive an expression for right angle triangular lamina. CO4 (04)

c) Describe the following with relevant examples: CO4 (06)

i) Dry Friction ii) Fluid Friction iii) Static Friction iv) Limiting Friction.

8. a) Two blocks A & B weighing 3 KN and 1.3 KN respectively are connected by a string over a frictionless pulley as shown in Fig. 8(a). Determine the minimum value of force T to generate an impending motion to the right. The coefficient of friction for the surface of contact for blocks A & B are 0.2 and 0.3 respectively. CO4 (08)

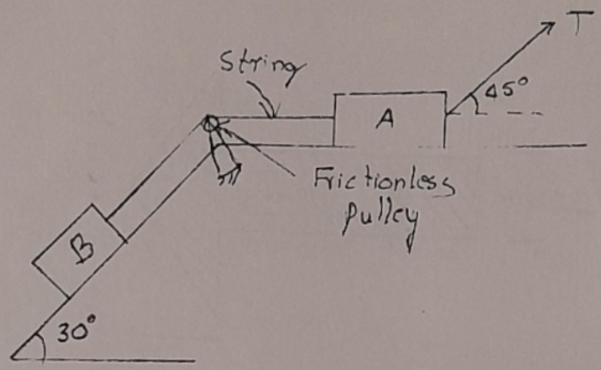

Figure 8a

b) Derive an expression for the centroid of semicircular lamina. CO4 (06)

c) Determine the centroid of the shaded area shown in Fig. 8(c). CO4 (06)

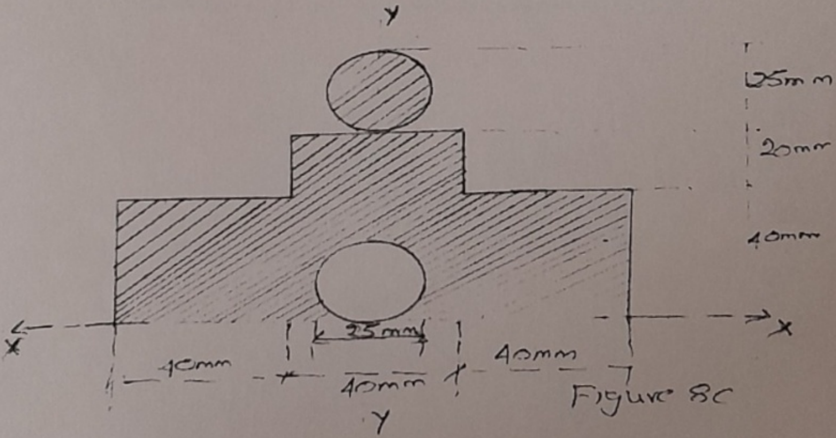

Figure 8c

UNIT- V

9. a) Compute the moment of inertia of the area shown in the Fig. 9(a) about the axis AB. CO5 (10)

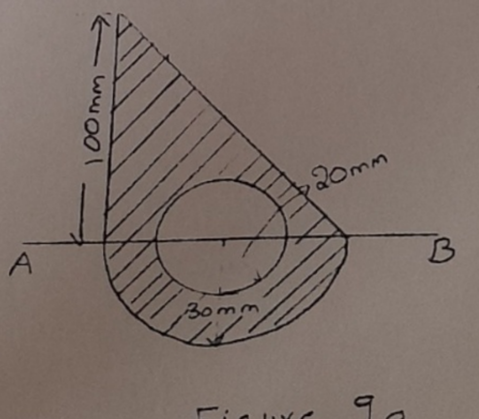

Figure 9a

b) State and prove parallel axis theorem. CO5 (06)

c) Define polar moment of inertia and radius of gyration. CO5 (04)

10. a) Derive an expression for the moment of inertia of a triangle about its base using method of integration. CO5 (08)

b) Determine the radius of gyration of the shaded area about an axis normal to the symmetrical axis. Refer Fig. 10(b). CO5 (12)

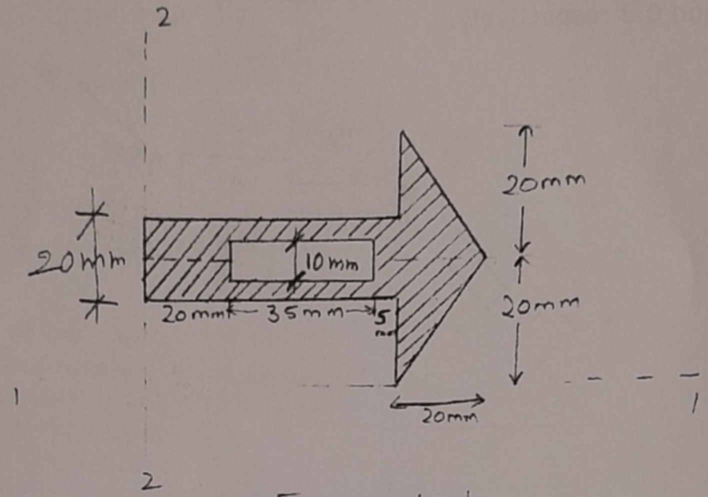

Figure 10 b

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